PLANNING EXPERIMENTS

When an experiment is conducted, it is important to control the variables that are possibly involved.

Variable: any changeable factor that may affect the results of an experiment.

Independent Variable: the variable that is deliberately ***changed*** to assess its effect.

Dependent variable: the variable being ***measured***.

Controlled variables: all variables that may affect the results and must be controlled.

Fair Test: being able to collect data that tests a hypothesis fairly. This is achieved by controlling variables correctly.

Reliability: do the tests more than once and average the results. this minimises any errors.

Example

A group of students tested the following hypothesis regarding table salt.

"Table salt will dissolve faster in hotter water."

They decided the following with regards to variables.

Independent variable: **(The one that is changed)**	Temperature (It will be changed by heating.)
Dependent variable: **(The one that is measured)**	Time to dissolve completely.
Controlled variables: **(Keep these constant)**	Same amount of water in a beaker. Same size of beaker. Same type of Bunsen flame or heat source. Same amount of salt being dissolved. Same amount of stirring (if used).